

Part 1: Conceptual Database Design

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1 Entity-Relationship (ER) Diagram

The conceptual database design was created using the standard Entity-Relationship (ER) notation described in the lectures. The diagram below models all entity sets, relationship sets, key constraints, participation constraints, and class hierarchies (Is-A relationships) as specified in the project requirements.

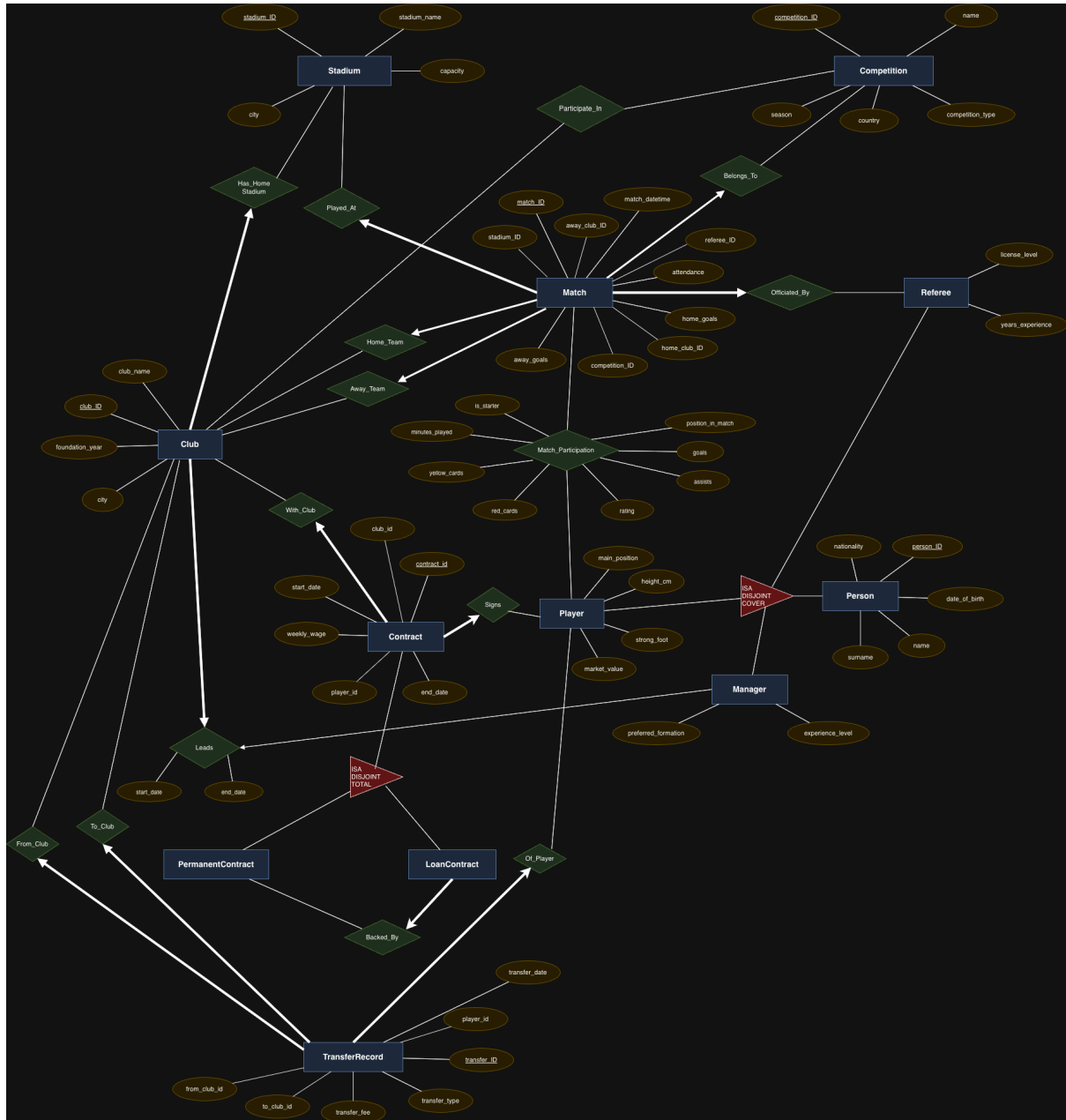


Figure 1: Conceptual ER Diagram for TransferDB

2 Legend

The graphical notation used in the ER diagram follows these conventions:

- **Blue rectangle:** Entity
- **Yellow oval:** Attribute
- **Green diamond:** Relationship
- **Red triangle:** ISA relationship
- **Underlined:** Primary Key
- **Bold arrow toward relation:** Total participation + Key constraint
- **Thin arrow toward relation:** Partial participation + Key constraint
- **Bold line toward relation:** Total participation + No key constraint
- **Thin line toward relation:** Partial participation + No key constraint

3 Constraints Not Expressible in ER Design

While the ER diagram captures the main entities, relationships, and structural constraints of the system, some business rules cannot be fully represented in standard ER notation. In particular, temporal constraints, inter-row dependencies, derived conditions, deletion/history-preservation requirements, and detailed domain validations must be enforced at the logical design stage through SQL `CHECK` constraints, triggers, or application-level logic.

1. The combination (`stadium_name`, `city`) must be unique.
2. The combination (`competition_name`, `season`) must be unique.
3. `attendance` must not exceed the capacity of the stadium in which the match is played.
4. No two matches may be scheduled at the same stadium if their start times are within 120 minutes of each other.
5. A club may not appear in two matches whose start times are within 120 minutes of each other.
6. A referee may not be assigned to two matches whose start times are within 120 minutes of each other.
7. At most 11 players may be marked as starters for the same club in a given match.
8. At most 23 players may be registered in the squad for the same club in a given match.
9. Suspension rules derived from disciplinary records cannot be represented directly in the ER diagram, including:
 - a red card causing suspension for the club's next match in the same competition,
 - accumulation of 5 yellow cards in a season causing suspension for the next match in the same competition.
10. Attribute-domain constraints such as positivity, non-negativity, bounded ranges, and enumerated-value restrictions cannot be fully captured in ER notation. Examples include:
 - `market_value` > 0
 - `height_cm` > 0
 - `years_experience` ≥ 0
 - `capacity` > 0
 - `weekly_wage` > 0
 - `transfer_fee` ≥ 0
 - `home_goals`, `away_goals` ≥ 0

- `minutes_played` $\in [0, 120]$
 - `yellow_cards` $\in \{0, 1, 2\}$
 - `rating` $\in [1.0, 10.0]$
11. The foundation year of a club must be a valid past year.
 12. The end date of a contract must be strictly later than its start date.
 13. A player may not hold two active contracts of the same type simultaneously.
 14. A loan contract is valid only if the player simultaneously has an active permanent contract with a different club.
 15. A player may be associated with at most two clubs at the same time: one through an active permanent contract and one through an active loan contract.
 16. A player with no active contract is considered a free agent; this is a temporal condition and cannot be explicitly enforced in the ER diagram.
 17. For a transfer record, `from_club_id` and `to_club_id` must refer to different clubs.
 18. Transfer-specific fee rules cannot be expressed in ER notation: a Free transfer must have `fee = 0`, while Purchase and Loan transfers may have positive fees.
 19. A Loan transfer is valid only if the player has an existing active permanent contract, and the parent club must match `from_club_id`.
 20. Manager–Club assignments involve temporal exclusivity: a manager cannot lead two clubs at the same time, and a club cannot have two managers at the same time.
 21. In a match, the home club and away club must be different clubs.
 22. A player cannot appear in the same match as both a home-team participant and an away-team participant.
 23. Receiving two yellow cards in a single match automatically implies a red card; this is a derived rule rather than a directly modeled ER constraint.
 24. The match result (Home Win / Away Win / Draw) is derived from `home_goals` and `away_goals`, and is therefore not stored as an independent attribute.
 25. Contract records are historical and must never be deleted, even after expiration.
 26. Transfer records are historical and must be permanently preserved, even after the player leaves a club.